

5.2 Permutations and combinations

Probability and Statistics 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Permutations count ordered selections $nPr = \frac{n!}{(n-r)!}$; combinations count unordered groups $nCr = \frac{n!}{r!(n-r)!}$. For arrangements with repeated identical objects divide by repeated factorials. Restrictions such as adjacency are often handled by blocks or complement.

Worked example: follow the reasoning

Number of unordered groups of 3 from 8 is ${}^8C_3 = 56$, while ordered selections are ${}^8P_3 = 336$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 How many different arrangements of letters LEVEL are there?

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Five positions with two L and two E [2]; $5!/(2!2!)=30$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.